DMBI CA2 Mini Project · D15A Group 09 · SDG 3 and SDG 11

BlackSpot Miner

Mining road collision data to locate severe-crash hotspots and prioritise road safety action

513,801
real collisions, 2021 to 2025

804
street-scale hotspots of fatal and serious crashes

107 of 109
rules held on unseen 2025

0.764
next-year site ROC-AUC, past counts 0.680

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A safety budget cannot fix every road, so where should it go?

1.19 million

road deaths a year worldwide (WHO, 2023)

1.72 lakh

road deaths in India in 2023 (MoRTH)

SDG 3.6

target on road traffic deaths and injuries

Where?

Which street-scale places concentrate severe collisions?
DBSCAN hotspots

Why?

Which conditions raise the chance of a death or serious injury (KSI)?
Apriori rules, K-Modes segments

What next?

Which sites should be treated first next year?
Classification and site ranking

Eight stages take raw records to a decision

1

Problem

Limited budget, where to act

2

Data

513,801 collisions

3

Preprocessing

Audit, clean, encode

4

Algorithm

DBSCAN, Apriori, K-Modes, classifiers

5

Results

Hotspots, rules, site scores

6

Evaluation

Unseen 2025 test year

7

Insights

Places and combinations

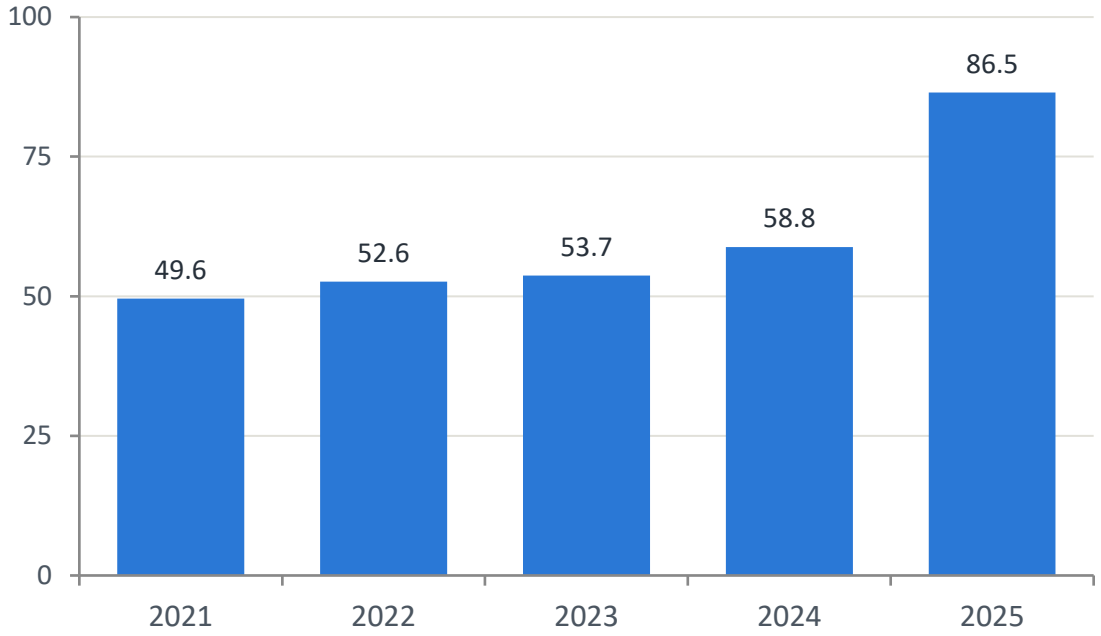
8

Decision

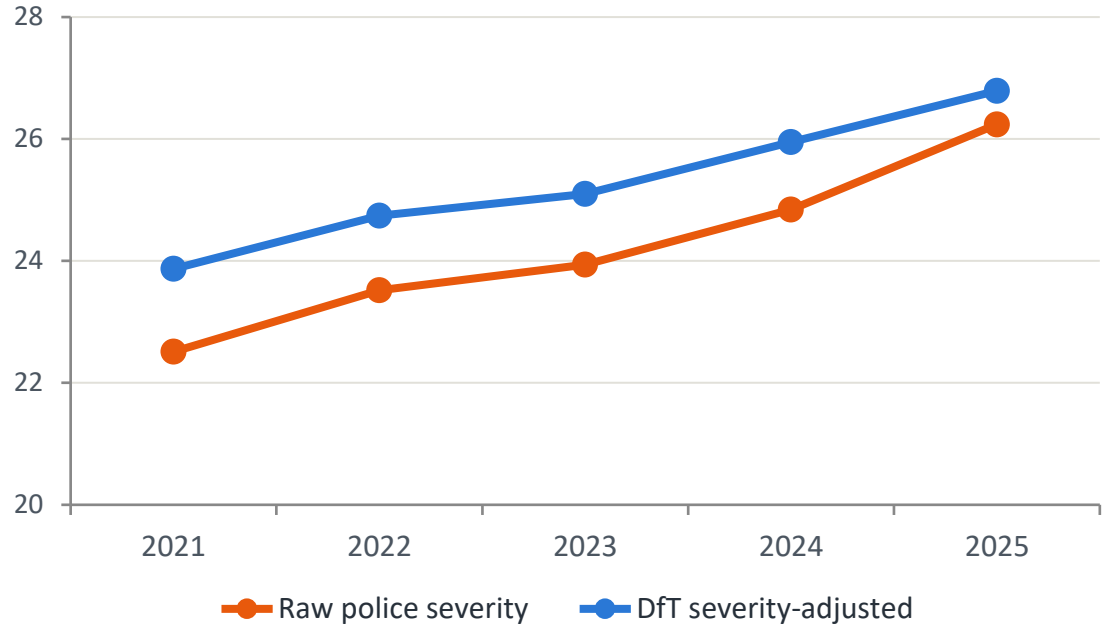
2026 priority list

A reporting change would have faked a trend, so we audited first

Collisions under injury-based reporting (%)



KSI share of all collisions (%)



Why it matters

15 of 44 police force codes changed system in five years. Raw KSI share goes from 22.51% to 26.24%. A model trained on it would partly learn a change of form, not a change in road risk.

Preprocessing made five years of records comparable

Integrate

Join 3 tables on collision_index. Derive flags: motorcycle, pedestrian hit, left the road, overturned.

Remove leakage

Drop number of casualties, police attendance and enhanced severity. All are known only after the outcome.

Clean

Remove 56 unusable rows. Drop fields that are 86.5% missing in 2025. Keep Unknown explicit.

Split by time

Train 221,193 rows on 2021 to 2024. Test 87,848 rows on 2025. No random split.

513,745

rows after cleaning, for location ranking

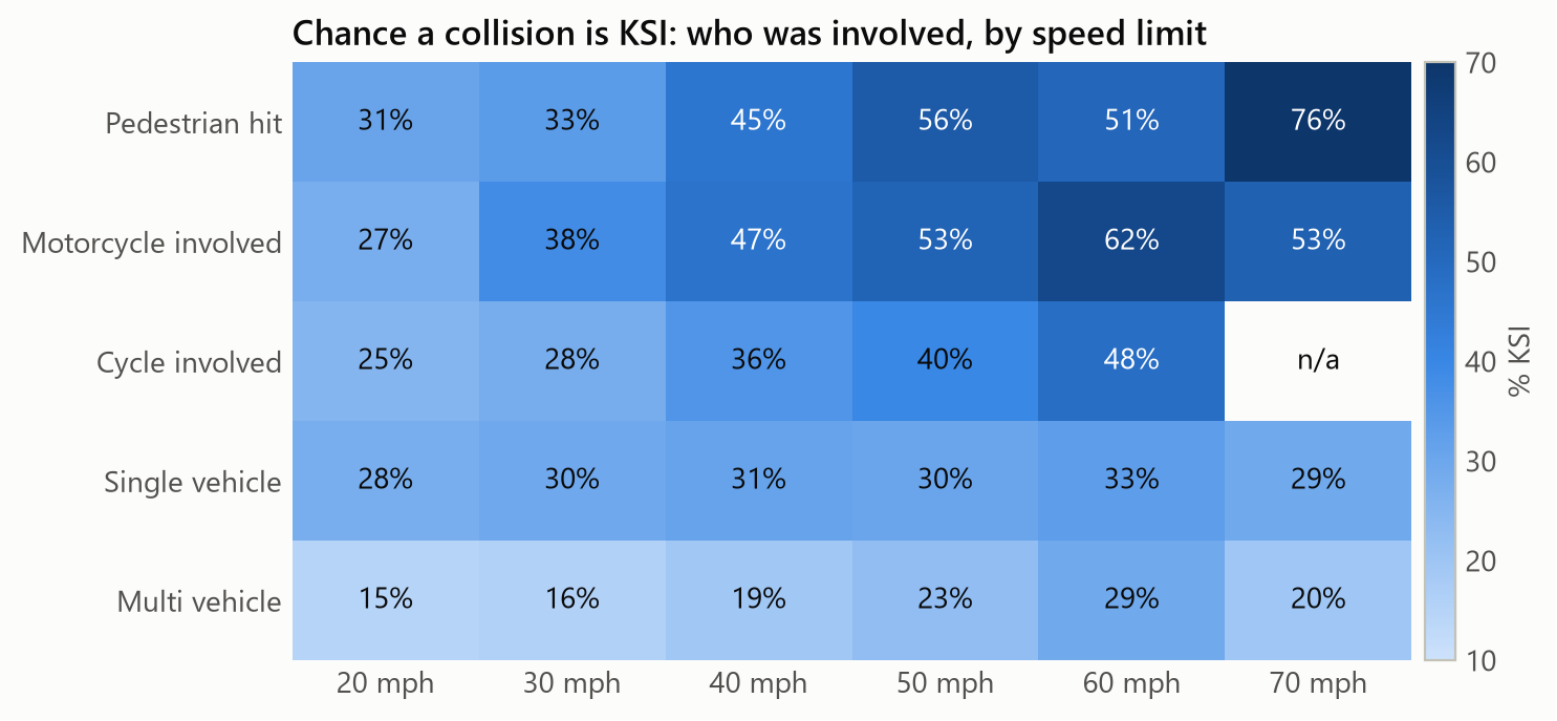
309,041

injury-based rows, for rules and severity

27.45%

of injury-based collisions are KSI

Speed hurts vulnerable road users far more than other crashes



Cells with under 200 collisions show n/a. Injury-based records only.

Most KSI collisions: Fri 15:00. Highest chance of KSI: Mon 04:00 (45.6%).

43.6%

of motorcycle collisions are KSI, against 27.4% overall

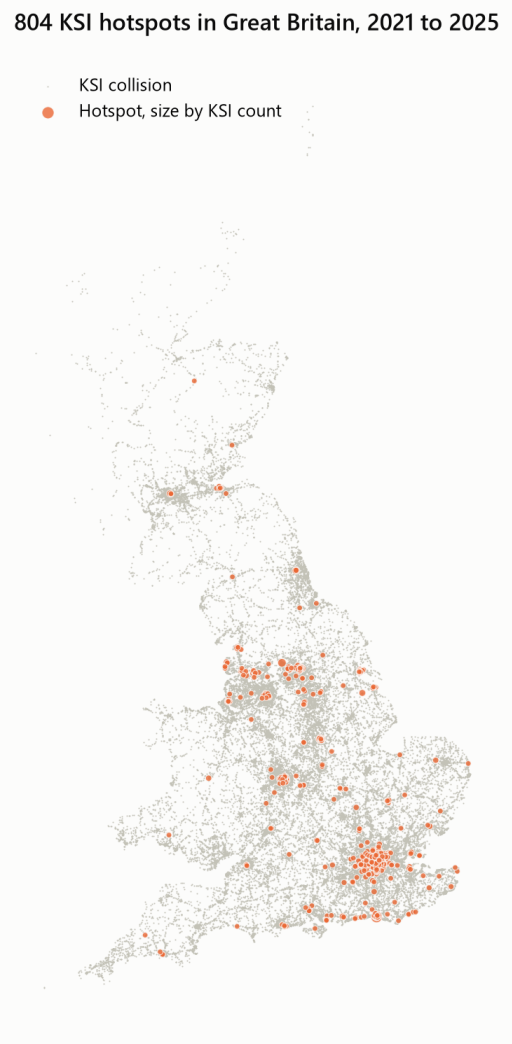
37.1%

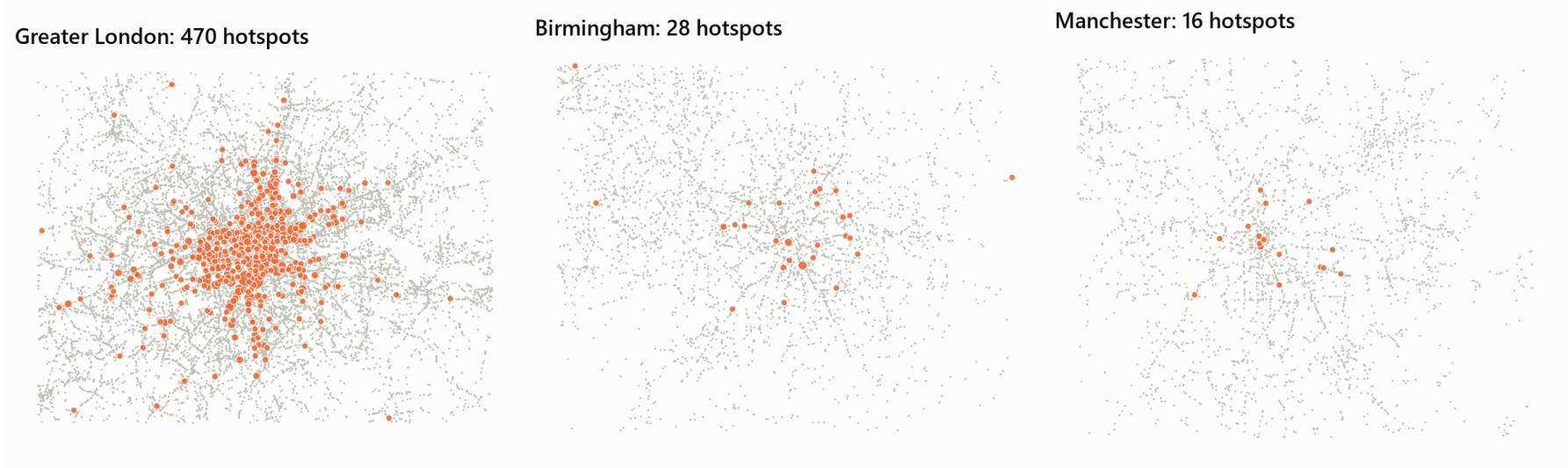
on 60 mph roads; motorways are 25.5%

36.9%

in darkness with no lighting; daylight is 26.3%

DBSCAN found 804 real hotspots, but most severe crashes are scattered





173 m

median hotspot span: a place an engineer can visit

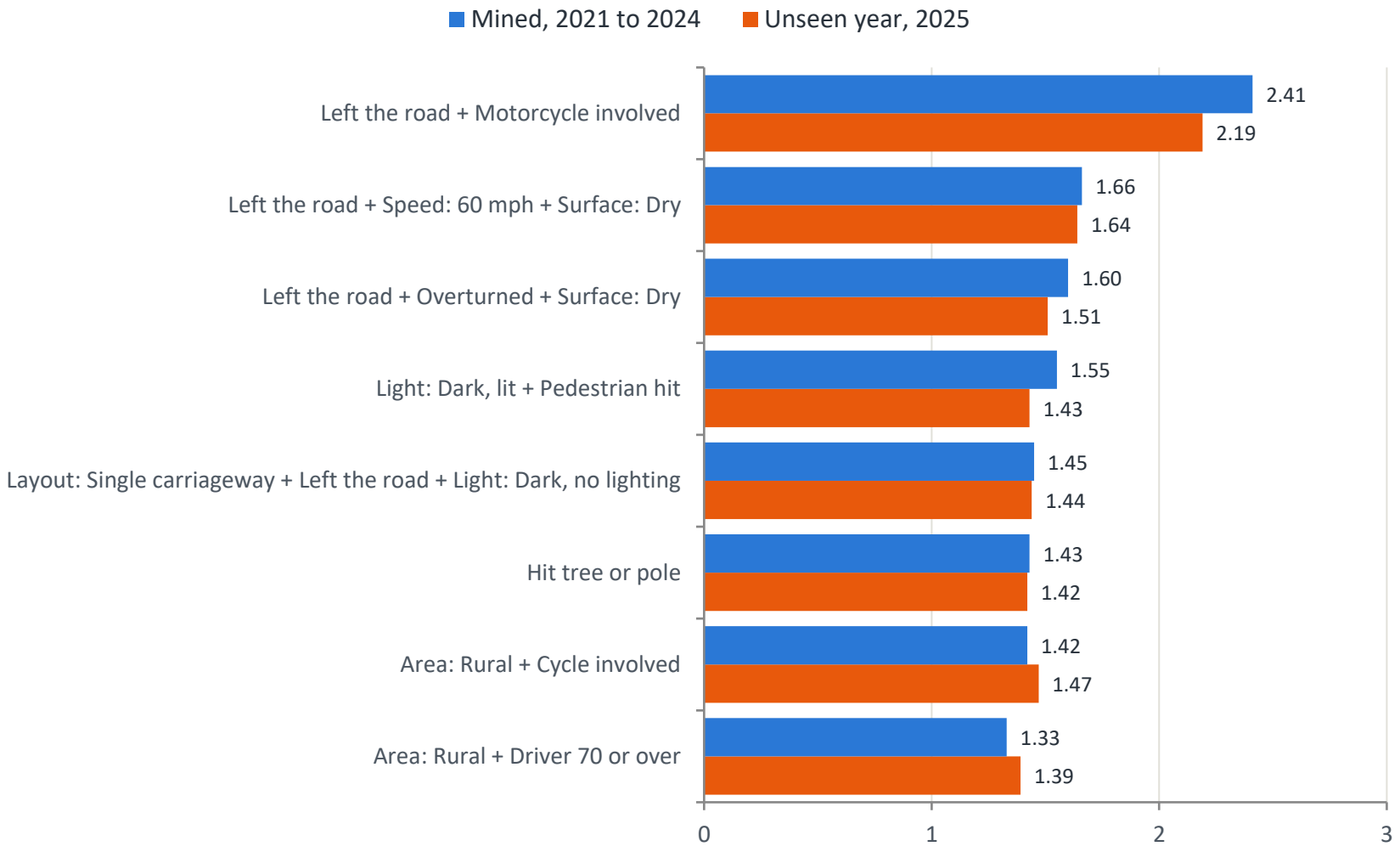
4.79×

2025 KSI per km², hotspots against random past sites

93.89%

of KSI collisions fall outside any hotspot

107 of 109 rules held on a year the mining never saw



0.946

correlation of lift between the two periods

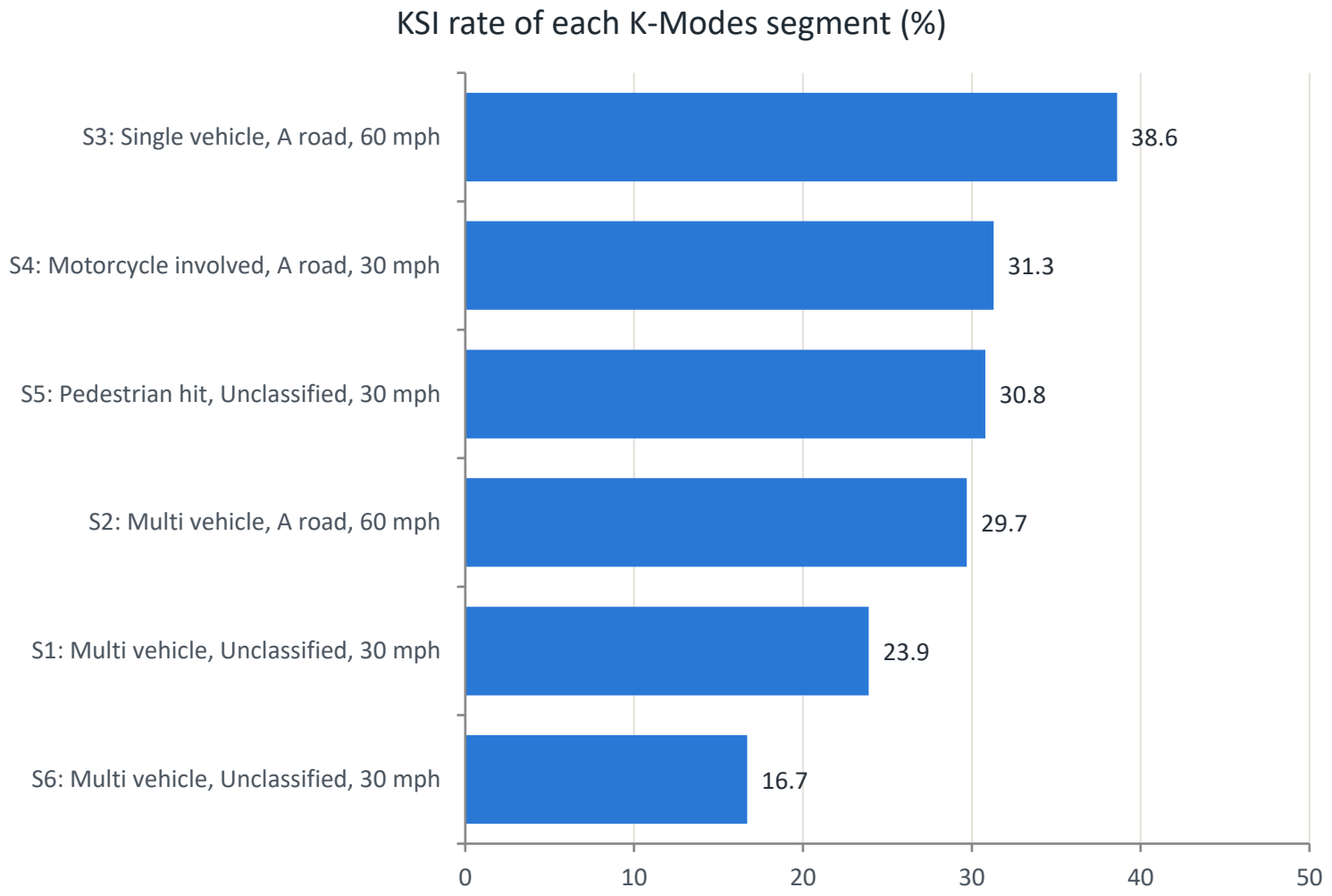
0.9%

median fall in lift on unseen data

Reading lift

2.0 means KSI is twice as likely as average. It shows association, not cause.

K-Modes segments confirm our reference paper's claim at scale



60

rules found only inside a segment

55

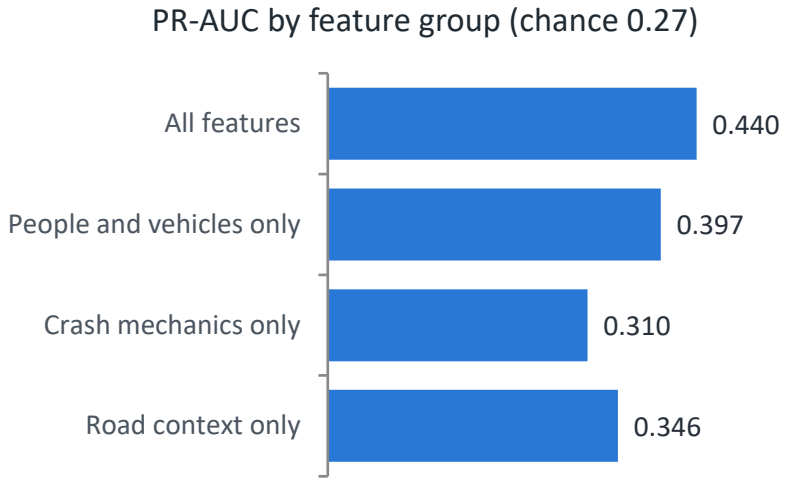
of those held on unseen 2025

Top hidden rule

Light: Dark, no lighting + Pedestrian hit: lift 1.78, 68.9% KSI inside S3.

Severity is only partly predictable, and that is a real finding

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	**56.9%**	**35.7%**	**74.4%**	**0.482**	**0.679**	**0.440**
AdaBoost	56.9%	35.4%	72.3%	0.475	0.670	0.426
Decision Tree	54.7%	34.5%	75.6%	0.474	0.665	0.422
Naive Bayes	54.4%	33.8%	71.7%	0.460	0.638	0.376



73.0%

accuracy from predicting Slight for everyone, catching no KSI collision

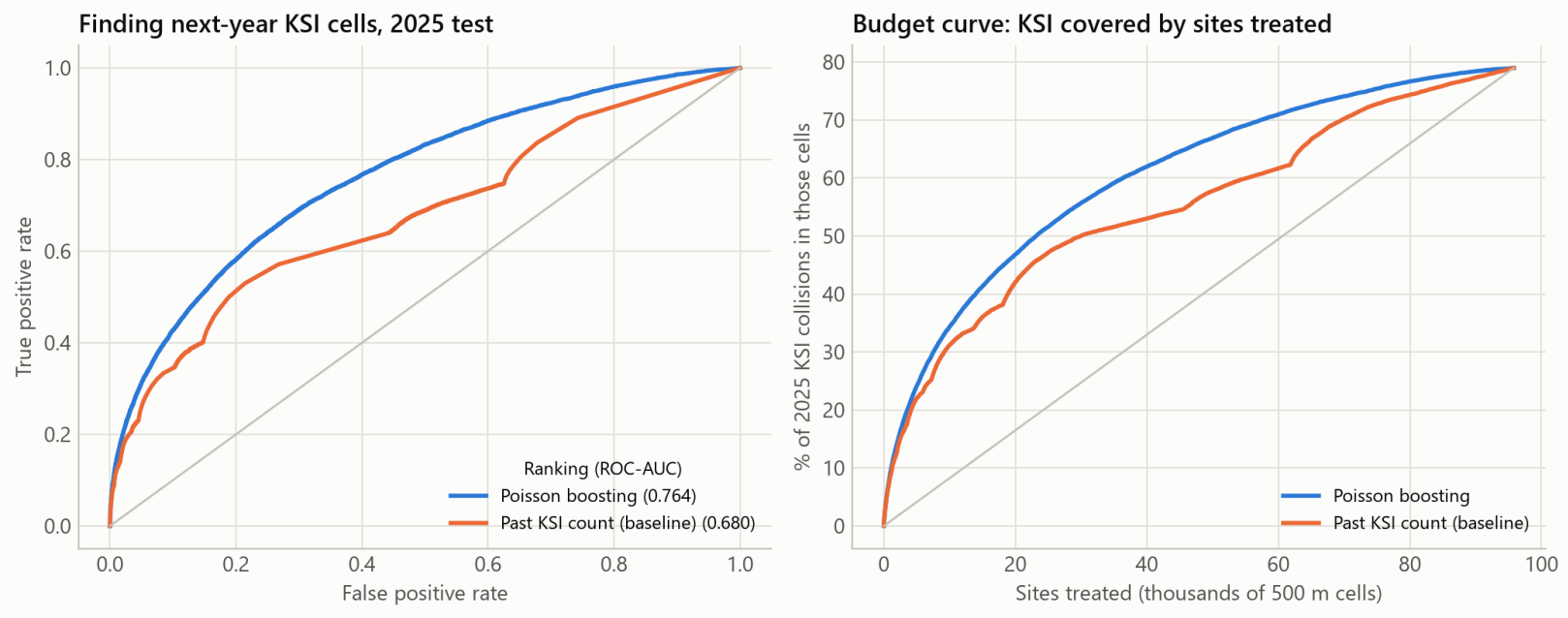
74.4%

of KSI collisions caught by Random Forest at the threshold set on 2024

Why not higher?

Impact speed, seat belt use and the health of the person hit are not recorded. Road context alone adds little.

Ranking next year's sites beats a past-count list



Sites treated	Past counts	Poisson boosting
500	5.4%	5.8%
1,000	8.6%	9.3%
5,000	22.0%	24.1%
10,000	31.2%	34.3%

+2.13 pts

KSI covered at 5,000 sites (95% interval 1.78 to 2.53)

-39.8%

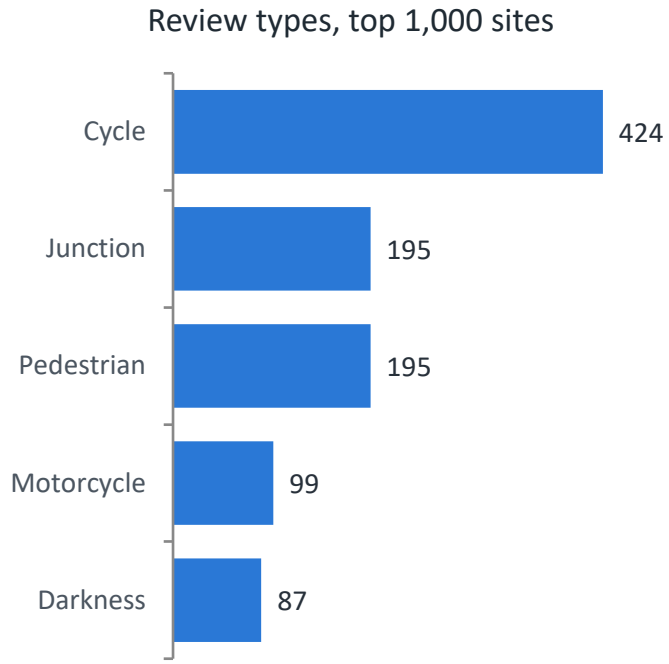
next-year fall in the 100 worst cells of 2024; the data records no treatment

78.98%

of 2025 KSI collisions happened in a cell with past collisions. The rest cannot be ranked by history.

The output is a 2026 priority list an engineer can act on

#	Road	Authority	KSI 2023-25	Suggested review
1	A3	City of London	23	Cycle separation and junction protection
2	A201	City of London	28	Cycle separation and junction protection
3	A201	Southwark	19	Cycle separation and junction protection
4	A3	Southwark	19	Cycle separation and junction protection
5	A401	Westminster	35	Pedestrian crossing and speed
6	A40	Westminster	21	Cycle separation and junction protection
7	A40	Westminster	29	Cycle separation and junction protection



How an authority uses it

Choose a budget, take that many sites from the list, read each site's pattern on the interactive map, and send an engineer to verify it.

98%

of the top 100 sites are near a DBSCAN hotspot

What we can claim, and what we cannot

Tested on unseen 2025

- 804 street-scale hotspots
- 107 of 109 rules held
- Site ranking beats past counts by 2.13 points at 5,000 sites
- The paper's clustering claim holds at scale

Limits

- No traffic volumes, so busy places rank high
- Association, not cause; no crash reduction claimed
- Police-reported injury collisions only
- Numbers describe Britain; the method transfers to India through eDAR

SDG 3.6 road deaths and injuries. **SDG 11** sustainable cities and safe transport.

Reference paper: Kumar, S. and Toshniwal, D. (2015). A data mining framework to analyze road accident data. Journal of Big Data, 2, 26.
link.springer.com/article/10.1186/s40537-015-0035-y
Data: Department for Transport, Road Safety Open Data, Open Government Licence v3.0.

Thank you. We are open for your questions.